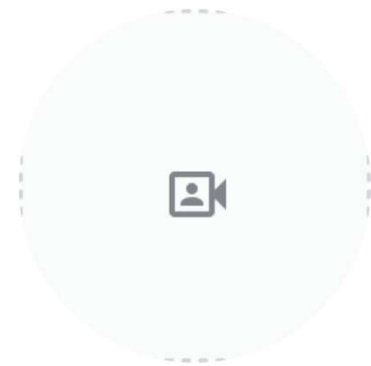


Cognitive psychology

MCA333D

COURSE INSTRUCTOR : Dr.TEGIL J JOHN



MISSION

CHRIST is a nurturing ground for an individual's holistic development to make effective contribution to

VISION

Excellence and Service

CORE VALUES

Faith in God | Moral Uprightness
Love of Fellow Beings

SYLLABUS

MCA 333D - Cognitive Psychology

Total Teaching Hours for Semester: 30

Max Marks:

100

Credits: 2

Course Objectives:

The course focuses on discussing and comprehending the fundamental cognitive processes that influence human behaviour and Human-AI interaction. The skills learn in this course will help in problem solving and decision making using cognitive tools. Understanding of human cognitive systems can be applied to the development of advanced Artificial Intelligence (AI) systems that are capable of learning from errors and improving the quality of our lives.

Course Outcome (CO):

Students will be able to:

CO1: Understand cognitive psychology, memory, language and communication.

CO2: Illustrate Problem solving, decision-making and learning

Unit 2: Memory and Learning

Unit 2: Memory and Learning

Teaching

Hours: 6

Models of memory and forgetting - Encoding, storage, and retrieval processes - Long-term memory structures and organization - Factors influencing memory performance

<https://www.slideshare.net/SureshababuG11/forgetting-definition-types-of-forgetting-theories-of-forgetting-ebbinghaus-forgetting-curve-causes-of-forgetting>

<https://nobaproject.com/modules/memory-encoding-storage-retrieval>

<https://www.simplypsychology.org/memory.html>

<https://psychology.pressbooks.tru.ca/chapter/10-2-long-term-memory-categories-and-structure/#:~:text=The%20structure%20of%20long%2Dterm,in%20common%20with%20each%20other>

<https://www.slideshare.net/SenTrisha/factor-affecting-memorypptx>

What is memory? Definition :

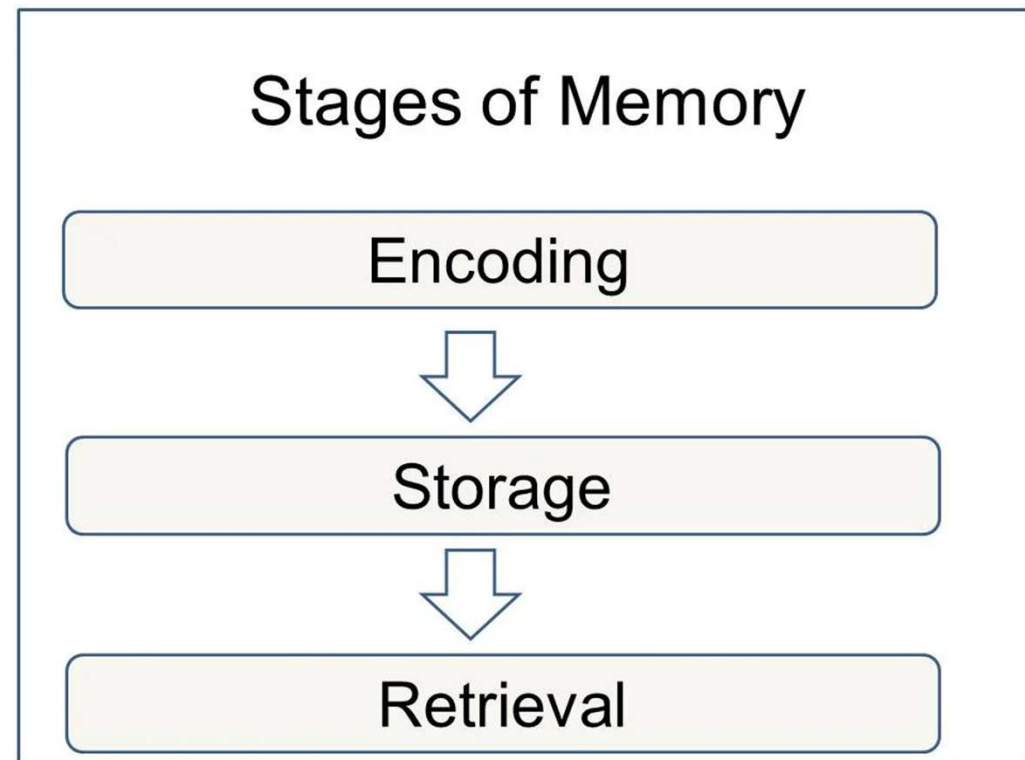
Our ability to **encode** store, retain and subsequently recall information and past experiences in the human brain. It is the sum total of what we remember, and gives us the capability to learn and adapt from previous experiences as well as to build relationships.

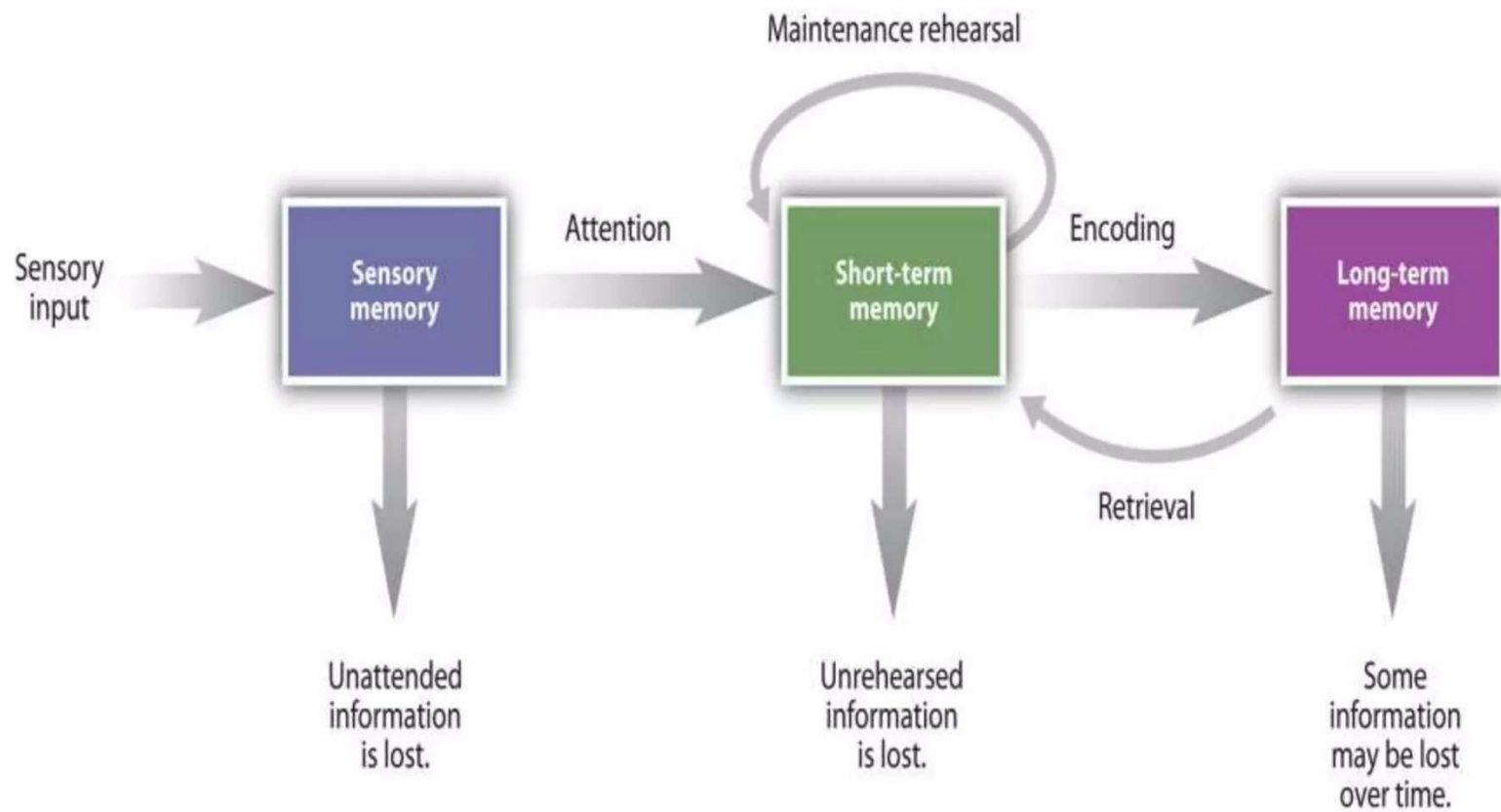
In more Physiological and neurological term, memory is, at its simplest , a set of encoded neural connections in the brain.

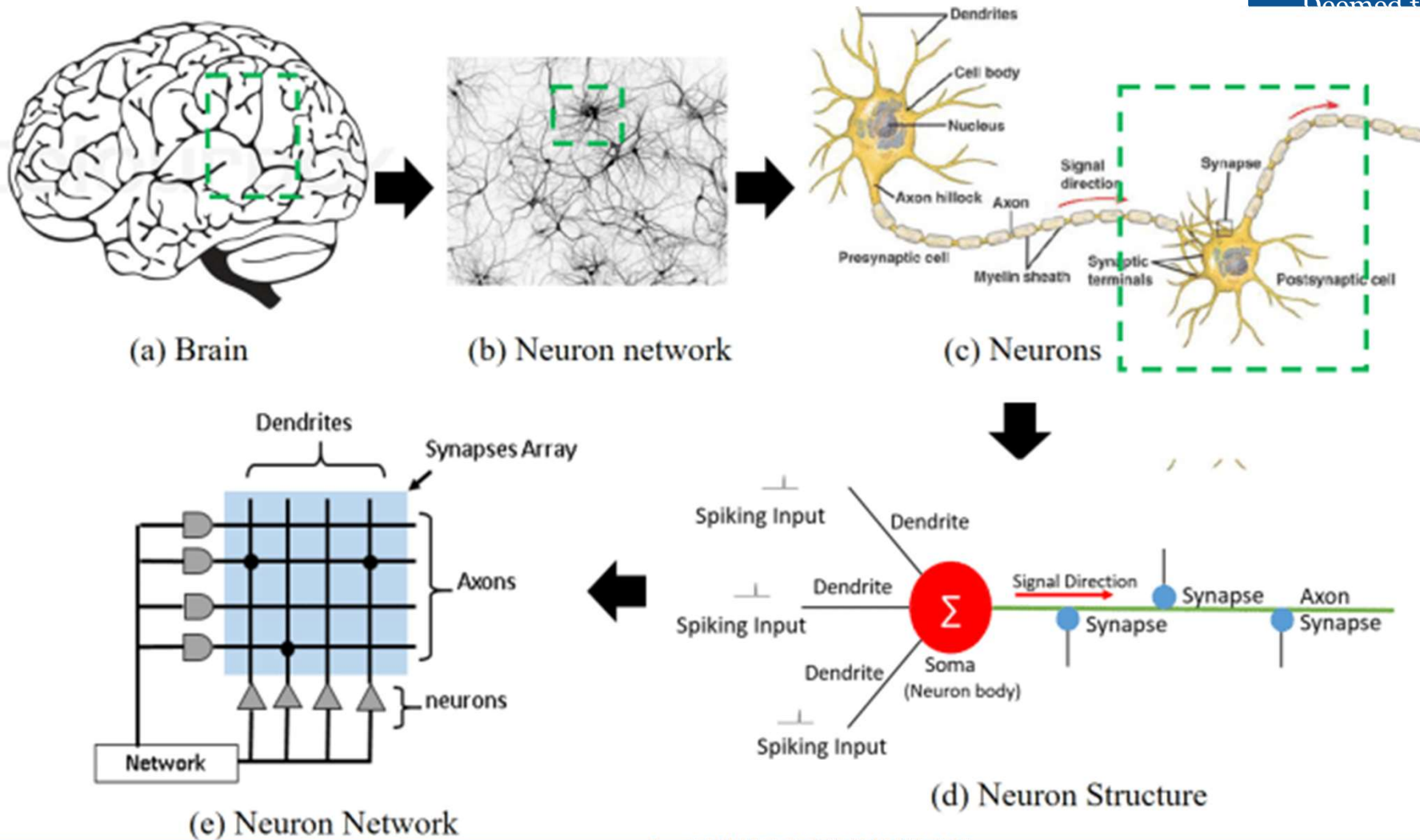


How does memory works?

- **Encoding:** the information gets into our brains in a way that allows it to be stored
- **Storage:** the information is held in a way that allows it to later be retrieved
- **Retrieval:** reactivating and recalling the information, producing it in a form similar to what was encoded



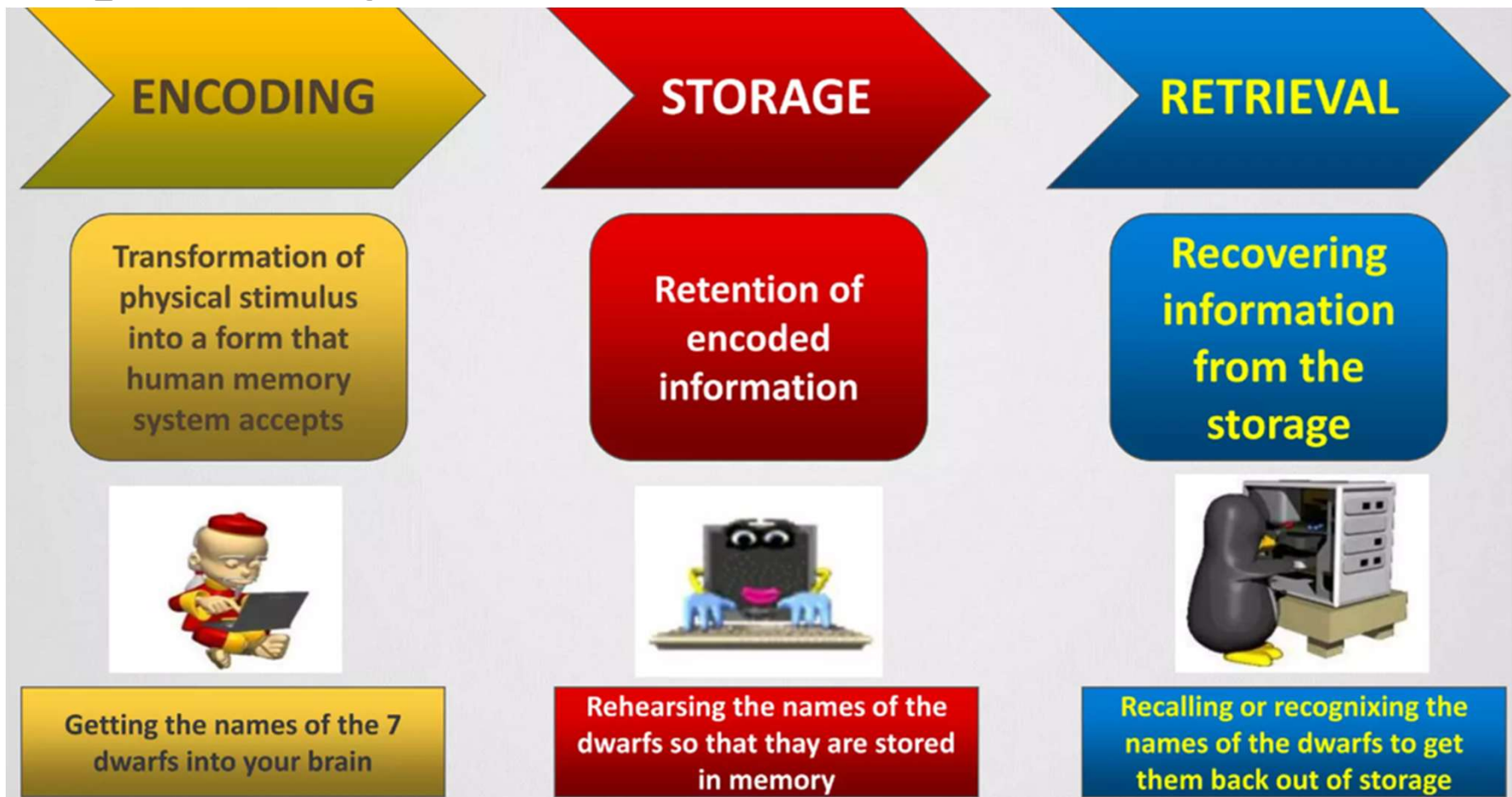




Concept of memory to neural network basics.

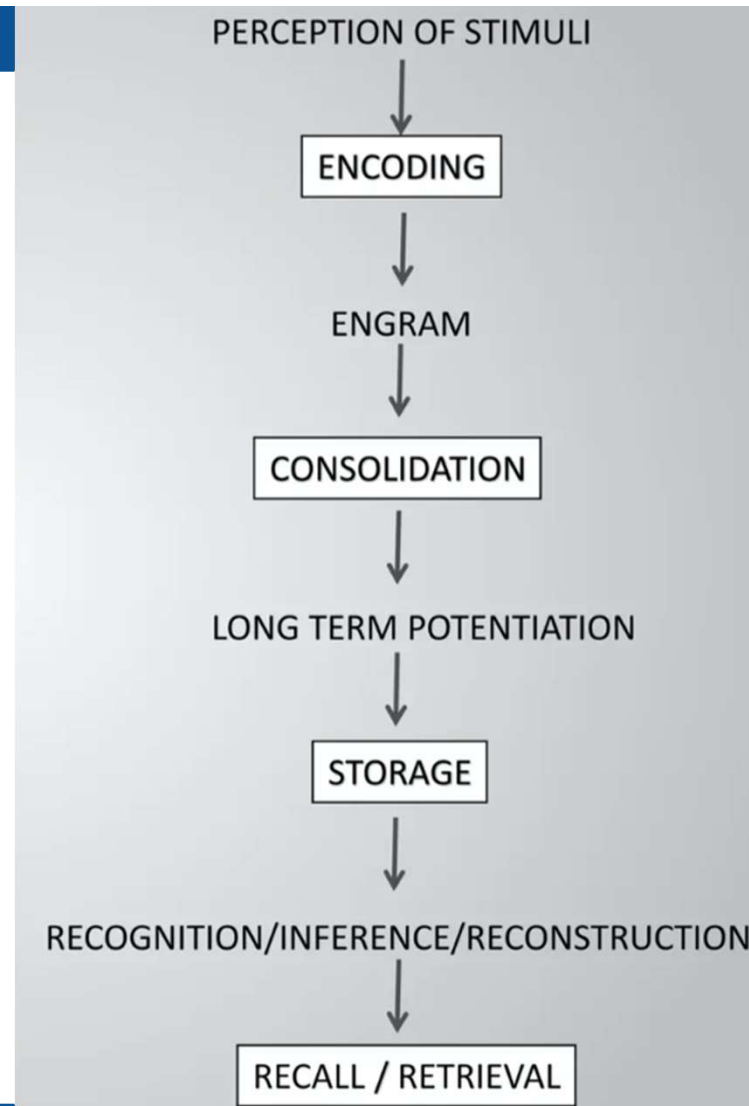
<https://youtu.be/6qS83wD29PY?feature=shared>

3 stage memory Process



Memory process diagram

An **engram** is a unit of **cognitive** information imprinted in a physical substance, theorized to be the means by which **memories** are stored as **biophysical** or **biochemical** changes in the brain or other **biological tissue**, in response to external stimuli.



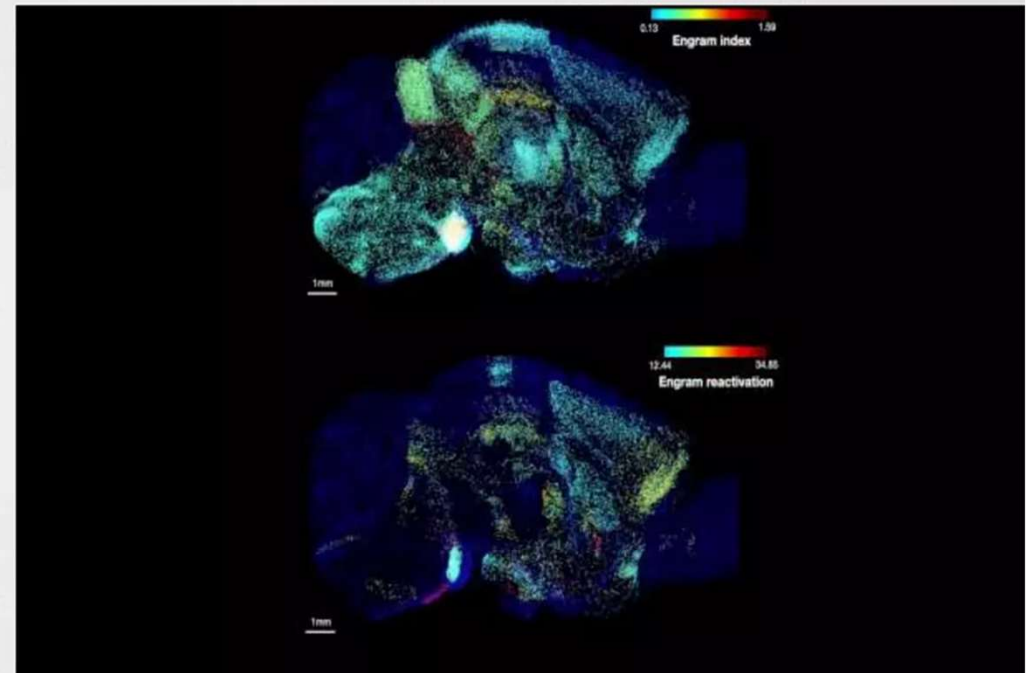
Encoding

Allows to perceive items of interest to be converted into a construct that can be stored within the brain and then recalled later from STM or LTM memory.

STEPS OF ENCODING

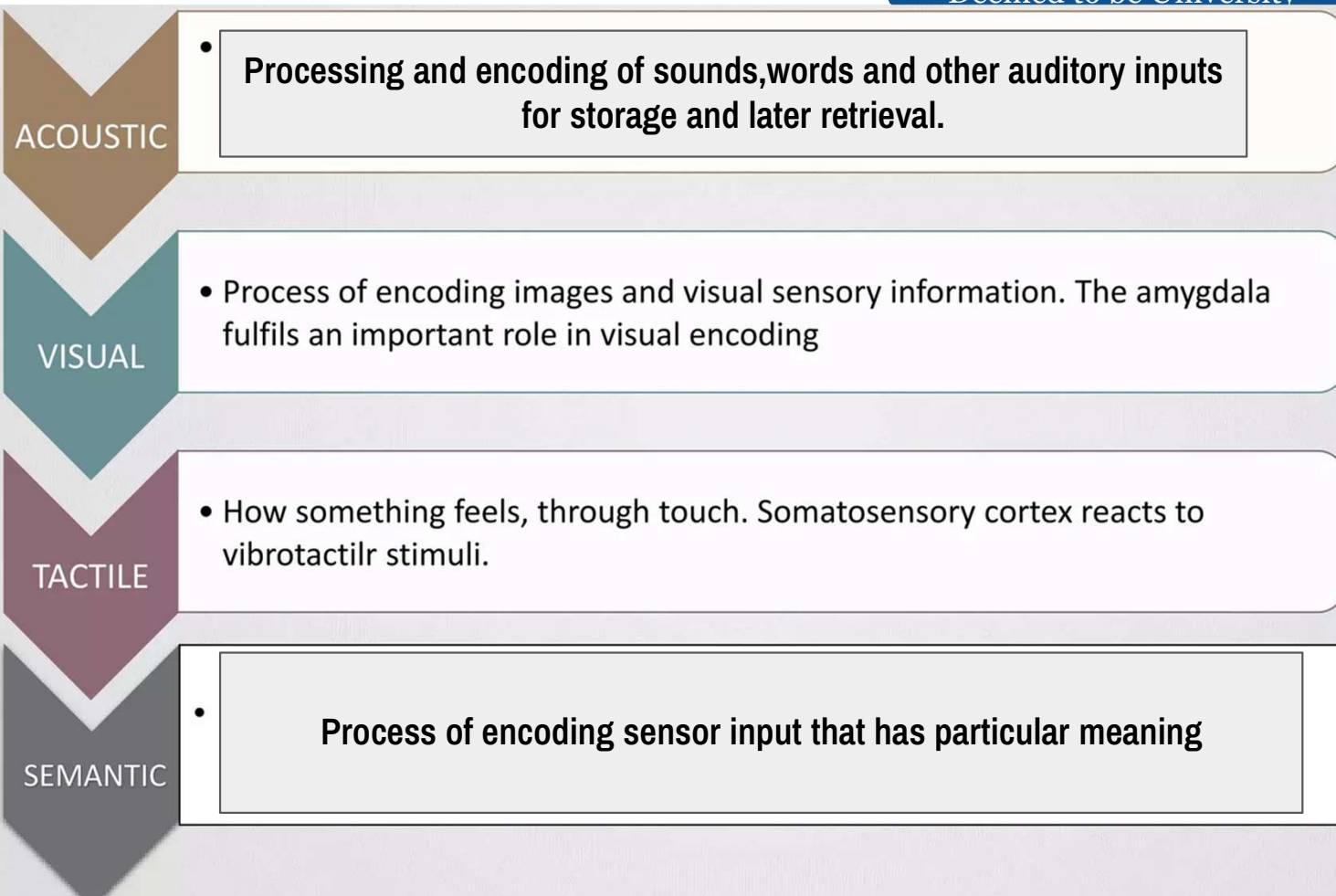
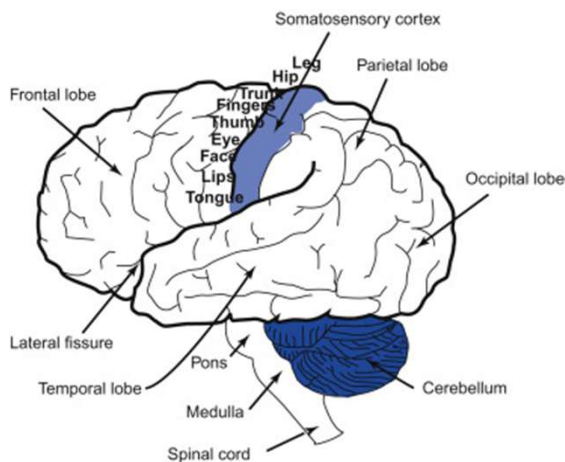
1) The formation of STM from the ultra short term sensory memory, followed by the conversion to the long term memory by the process of memory consolidation.

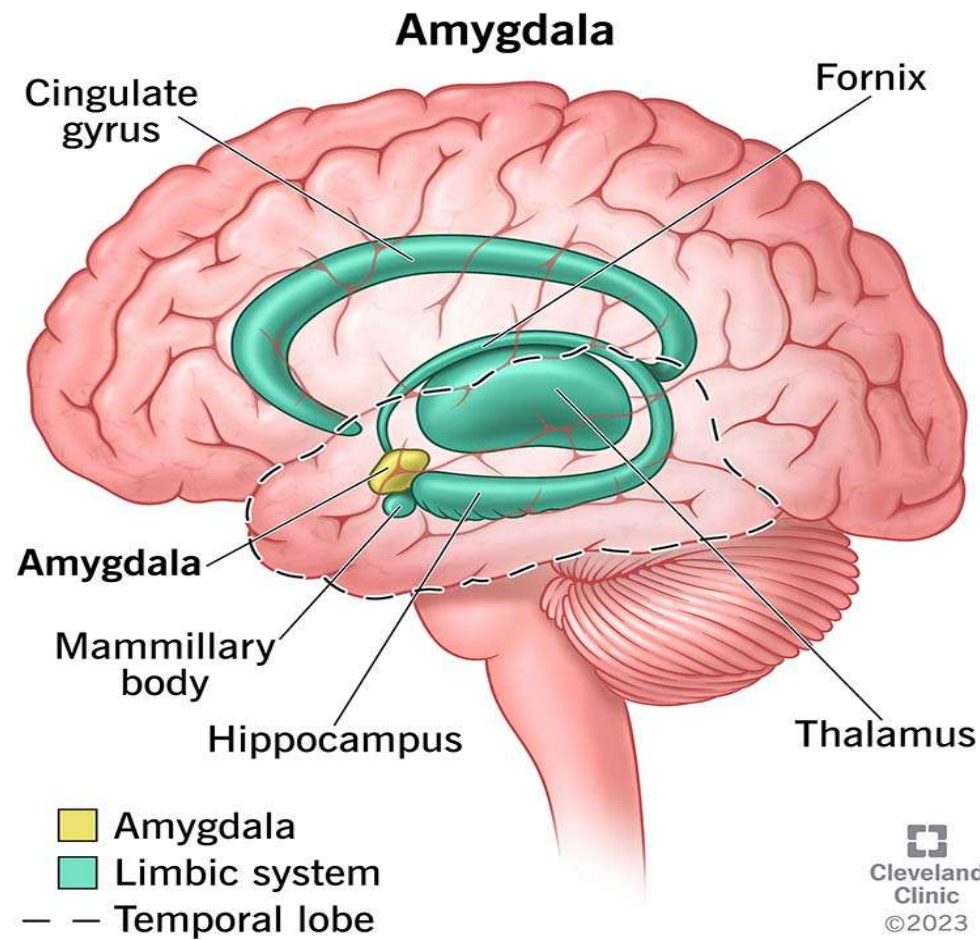
2) Engram - Hypothetical biophysical or biochemical change in the neurons of the brain, hypothetical in the respect that no one has ever actually seen, or even proved the existence of such a construct.



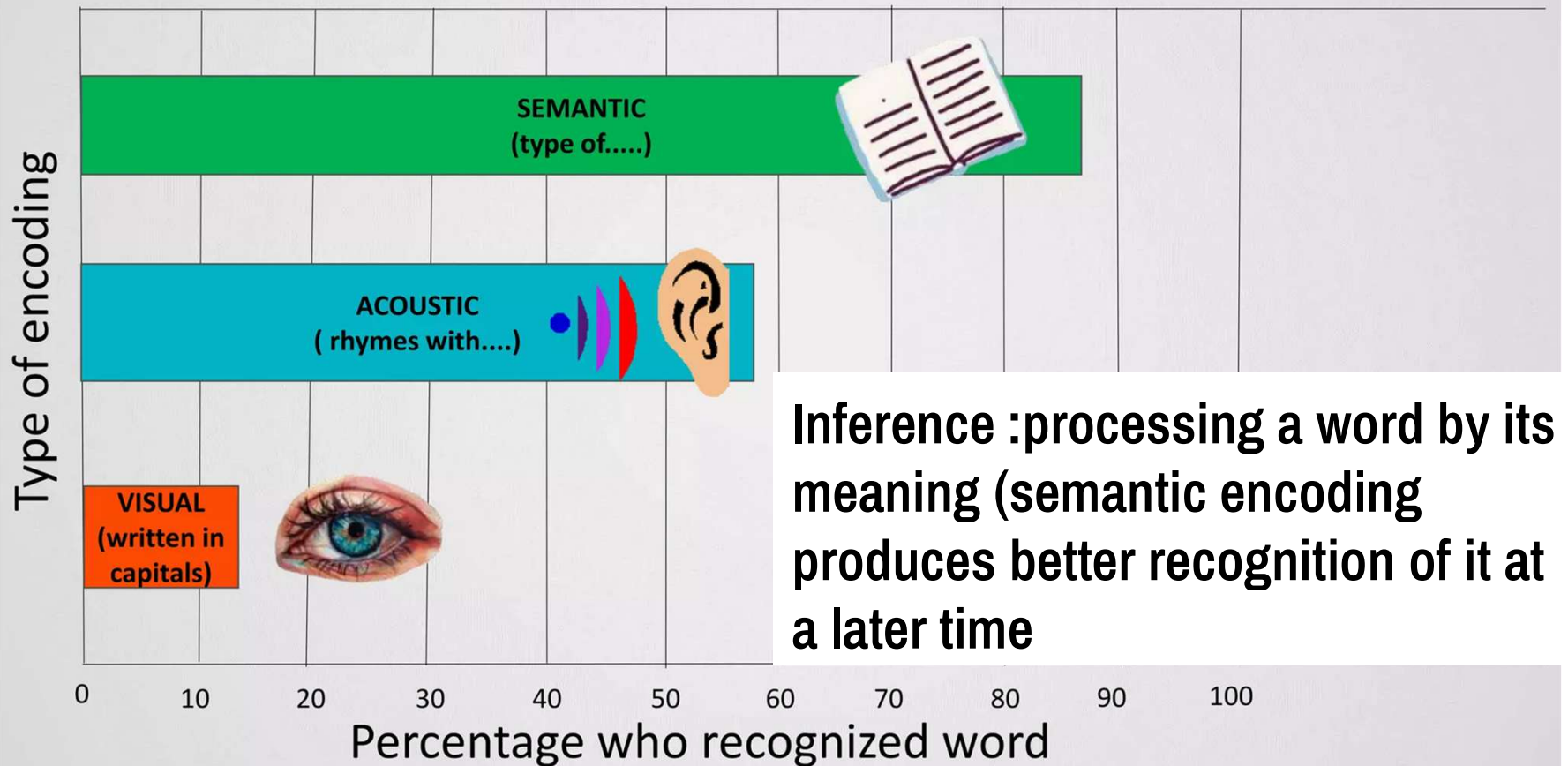
Many brain regions found likely to be involved in encoding a memory (top) were also found to be involved in recall upon reactivation (bottom). Credit: Tonegawa Lab/MIT Picower Institute

Types of encoding



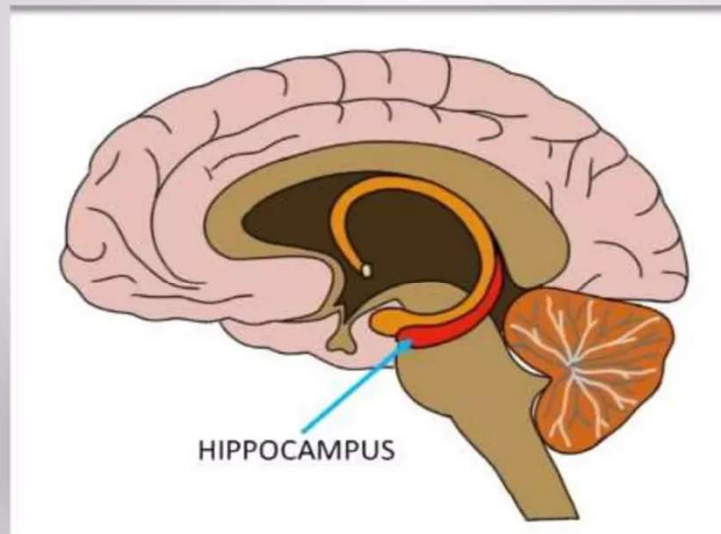


ENCODING-LEVEL OF PROCESSING



NEURO-BIOLOGY OF ENCODING

Process of laying down a memory begin with attention, which is regulated by thalamus and fontal lobe, in which a memorable event causes neurons to fire more frequently making the experience more intense and increasing the likelihood that the event is encoded as a memory. Emotion increases attention in the amygdala. The perceived sensation are decoded in the various sensory areas of the cortex. Then combined in the hippocampus into one single experience. Hyppocampus analyzing these inputs and ultimately deciding if they will be committed to long-term memory.



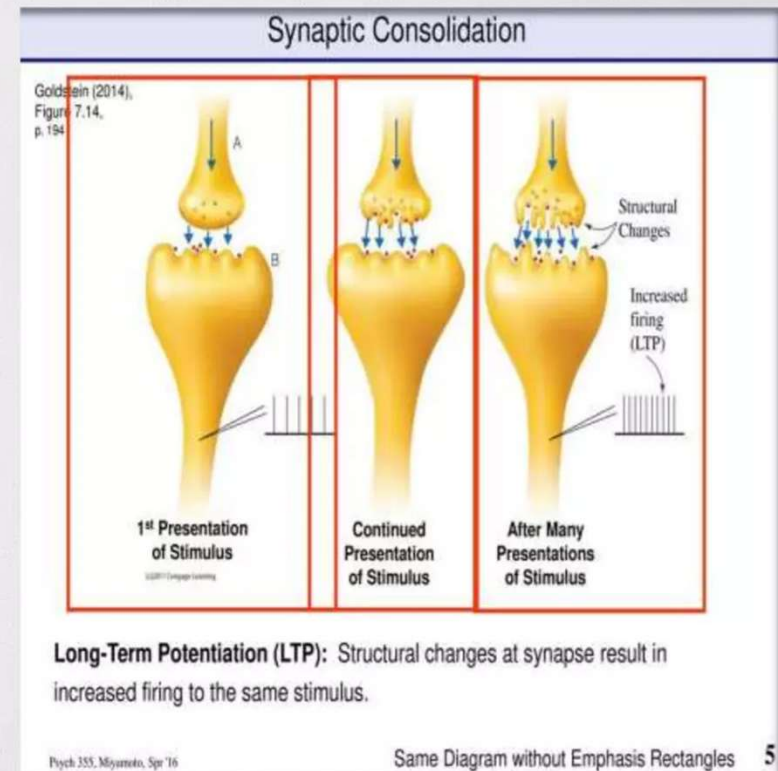
MEMORY CONSOLIDATION

Process of stabilising a memory trace after the initial acquisition.

2 Process -

Synaptic Consolidation - The first few hours after learning/encoding.

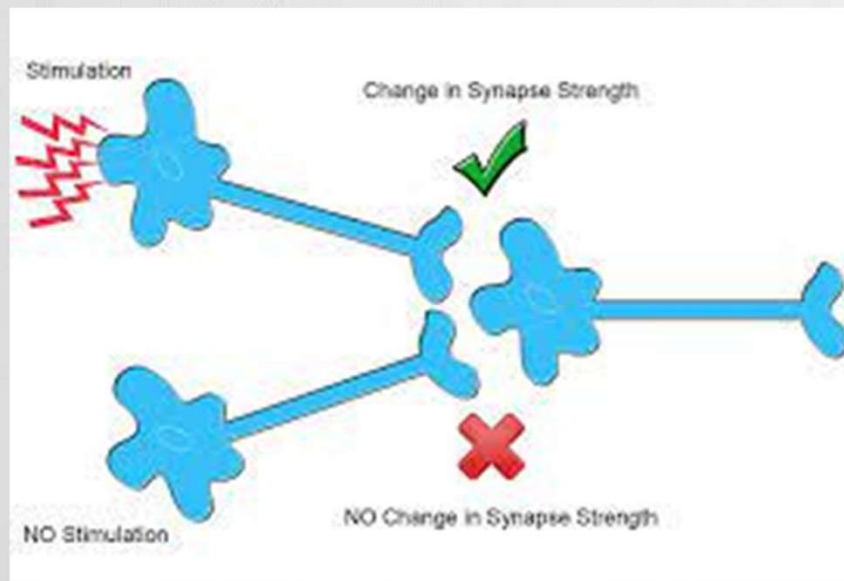
System Consolodation- Hippocampus-dependent memory become independent of the hippocampus over a period of weeks to years. The consolodation utilises a phenomenon called LT potentiation, which allows synamse to increase in strength as increasing number of signals are transmitted between 2 neurons. potentiation is the process by which synchronous firing of neurons makes those neurons more inclined to fire together in the future.



cont.....

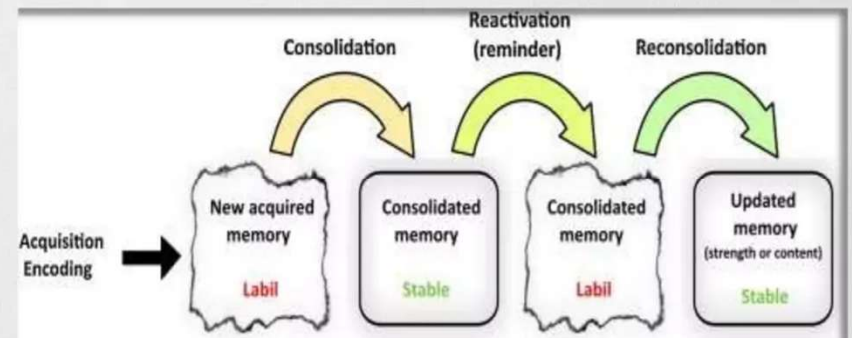
Synaptic Plasticity

The ability of the connection, or synapse between 2 neurons to change in strength and for lasting changes to occur in the efficiency of synaptic transmission.



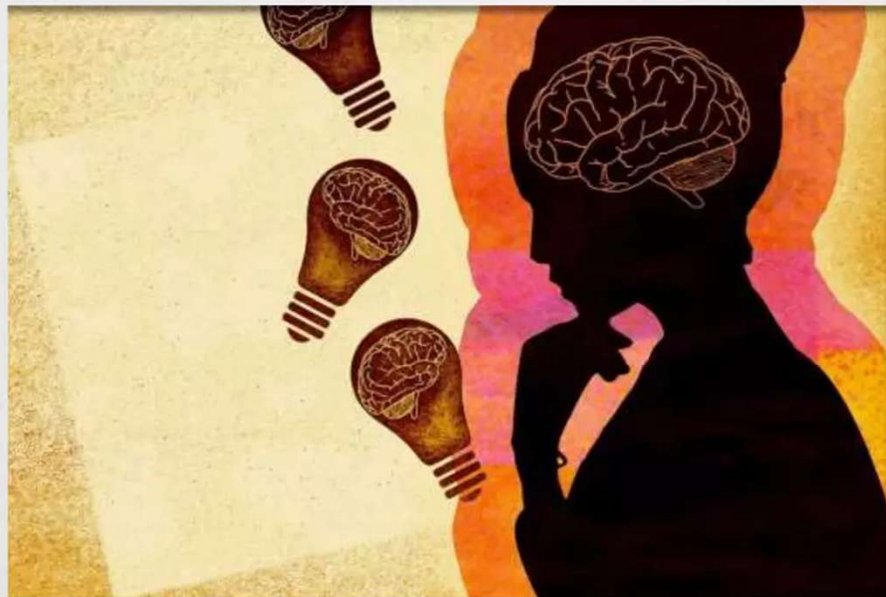
Memory reconsolidation

The process of previously consolidated memories being recalled and then actively consolidated all over again, in order to maintain, strengthen and modify memories that are already stored in the LTM.



MEMORY STORAGE

Passive process of retaining information in the brain. LTM are stored widely distributed throughout cortex. It is an ongoing process of reclassification resulting from continuous changes in our neural pathways and parallel processing of information in our brain.



Retrieval

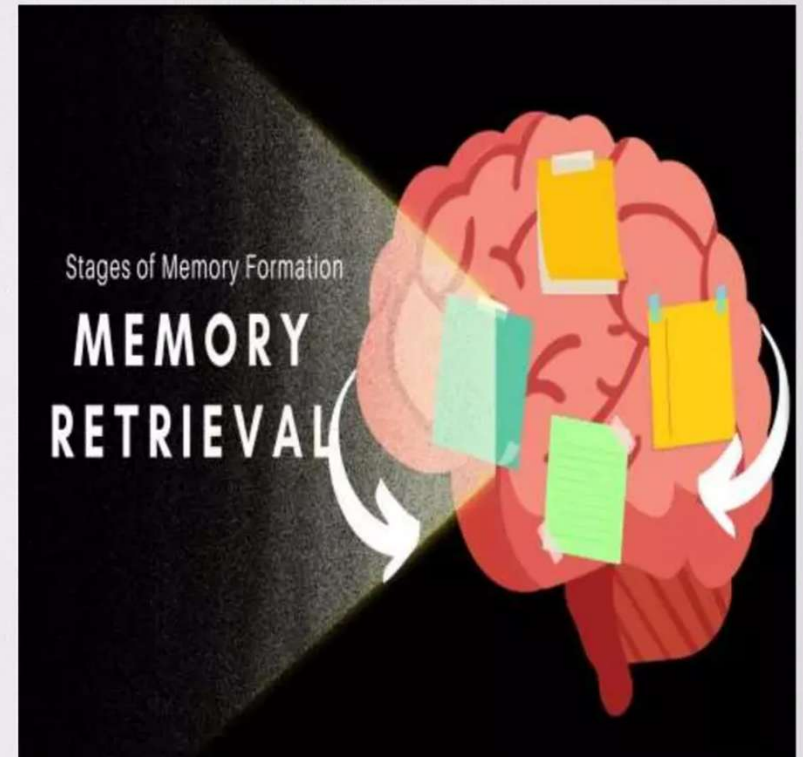
The subsequent re-accessing of events and information from the past.

Astounding-Efficiency of human memory recall.

2 main methods of accessing memory-

Recognition-An event or physical object with one previously experienced or encountered and involved a process of comparison of information with memory.

Recall- Remembering a fact, event or object that is not currently physically present.



TYPES OF RECALL

FREE RECALL- The process in which a person is given a list of items to remember and then is asked to recall them in any order.

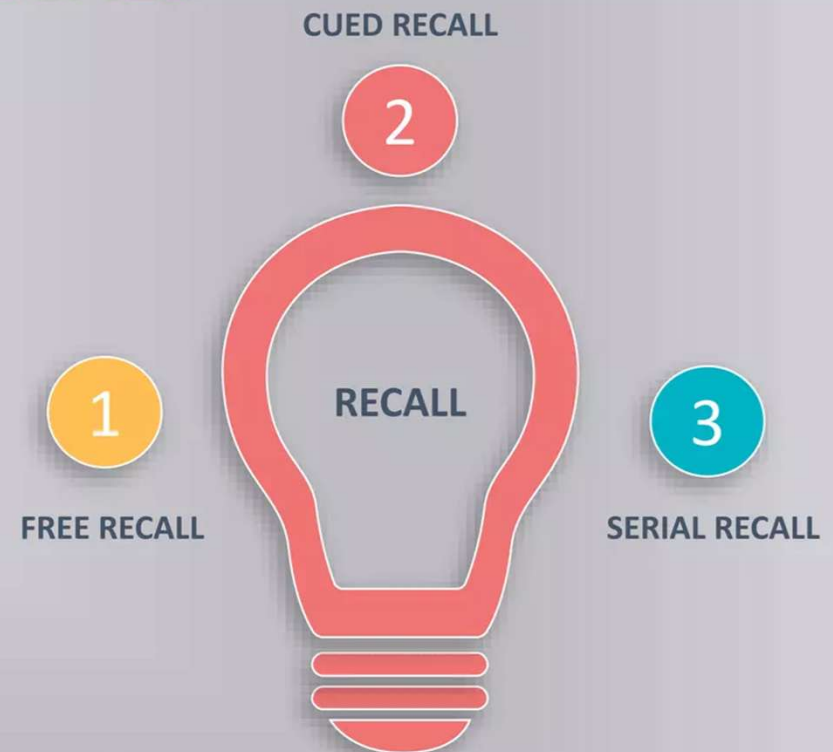
3 types- **Primacy effect-** Process recalls items presented at the beginning of the list earlier and more often.

Recency effect- When prson recalls items presented at the end of the list earlier and more often.

Contiguity effect- Marked tendency for items from neighbouring positions in the list to be recalled successively.

CUED RECALL- The process in which a person is given a list of items to remember and then tested with the use of cues and guides.

SERIAL RECALL- Ability to recall items or events in order in which they occurred, whether chronical events in our autobiographical memories or the order of the different parts of sentence in order to make sense of them.



Memory recall activity

View for 2 min .

Break for 2 min

Recall

How many you can ??

Recall the first row in order



FORGETTING

- Forgetting is the loss of the ability to recall or recognize something learned earlier.
- It is the failure of the individual to revive in consciousness an idea or group of ideas without the help of the original stimulus.

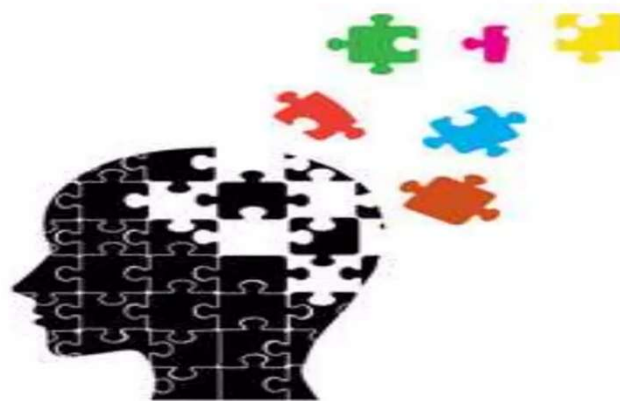


CHRIST
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Definition of Forgetting

- **Munn:** “Forgetting is the loss permanent or temporary of the ability to recall or recognize something learnt earlier”
- **Drever:** “Forgetting means failure at any time to recall an experience, when attempting to do so or perform an action previously done.



Kinds of Forgetting

- **Passive or Natural Forgetting** :
In this, there is no intention of forgetting on the part of the individual. For example, we forget many things with time lapse.
- **Active or morbid forgetting** : In this, the individual deliberately tries something. For example, we make deliberate efforts to forget the bitter experiences of life.



The Forgetting Curve



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- Curve of forgetting is a line-graph picturing the decline of retention with lapse of time.
- Forgetting increases with lapse of time, or in other words retention diminishes with time.
- It has also been noticed that the rate of forgetting is very initially and then diminished gradually over time.
- The curve obtained on the graph paper by plotting the amount forgotten as a function of time is named as curve of forgetting.
- The curve shows a rapid fall immediately after learning, and a gradual flattening out with increase of time interval.

Theories of Forgetting

There are four major theories of forgetting they are :

- **Decay Theory** : Which states that time alone causes memory traces to fade. Memory fades through disuse.
- **Interference Theory** : Which suggests that other memories interfere with remembering. There is interference from competing materials.



Decay theory

Interference theory

**Motivated
forgetting theory**

**Retrieval failure
theory**

Theories of Forgetting

- **Motivated Forgetting Theory** : Which suggests that we forget information that is unpleasant or threatening.
- **Retrieval Failure Theory** : Which propose that information memory must be distorted when we attempt to recall it. The right retrieval cue is not used.



Decay theory

Interference theory

Motivated forgetting theory

Retrieval failure theory

Example and discussion

Decay Theory:

- Example: Suppose you learn a new phone number but never use it or think about it again. Over time, you might forget the number due to decay

Inference Theory:

- Example: When you try to recall someone's name, but you keep thinking of similar names or associations, those interfering memories may prevent you from remembering the correct name.

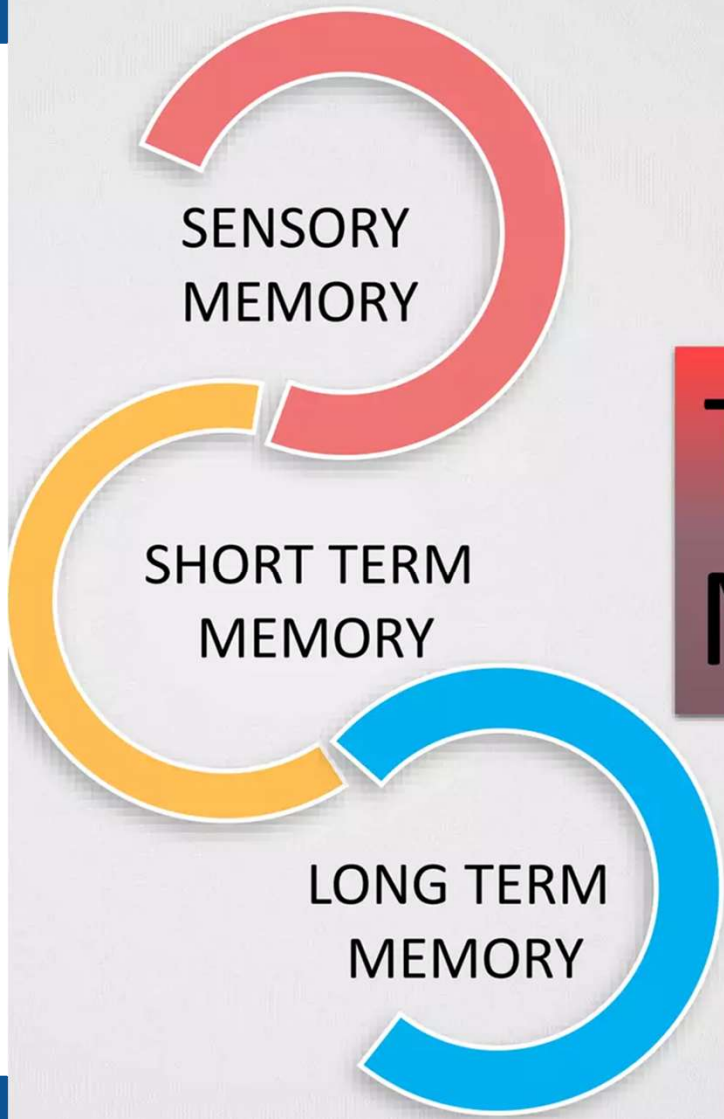
Motivated Forgetting Theory:

- Example: Someone may forget a traumatic event from their childhood because remembering it causes them distress. Alternatively, someone might forget to study for an exam deliberately because they fear failure.

Retrieval Failure Theory:

- Example: You might forget where you parked your car in a large parking lot. This isn't because the memory has decayed or been lost; rather, it's because the environmental cues (like landmarks or visual reminders) needed to recall the memory are absent at that moment.

TYPES OF MEMORY



The diagram illustrates the flow of information through three stages of memory. It features three large, overlapping, semi-circular arrows arranged in a vertical sequence. The top arrow is red and points downwards. The middle arrow is yellow and points downwards. The bottom arrow is blue and points downwards. The text 'SENSORY MEMORY' is centered within the red arrow, 'SHORT TERM MEMORY' is centered within the yellow arrow, and 'LONG TERM MEMORY' is centered within the blue arrow. The entire diagram is set against a light gray background with a subtle grid pattern.

SENSORY
MEMORY

SHORT TERM
MEMORY

LONG TERM
MEMORY

Case study read:

<https://ilslearningcorner.com/2016-04-types-of-memory-studies-show-working-memory-tied-to-your-childs-intelligence/>

THREE TYPES OF MEMORY

Sensory memory –Initial process that receives and holds environmental information in its raw form for a brief period of time, from an instant to several seconds

Short-term memory –Also called working memory; refers to another process that can hold only a limited amount of information an average of seven items, from 2 to 30 seconds

Long-term memory –Process of storing almost unlimited amounts of information over long periods of time

THREE TYPES OF MEMORY (cont.)

Memory processes

–Sensory memory

If we

don't pay attention, information is forgotten

pay attention, information is automatically transferred into short-term memory

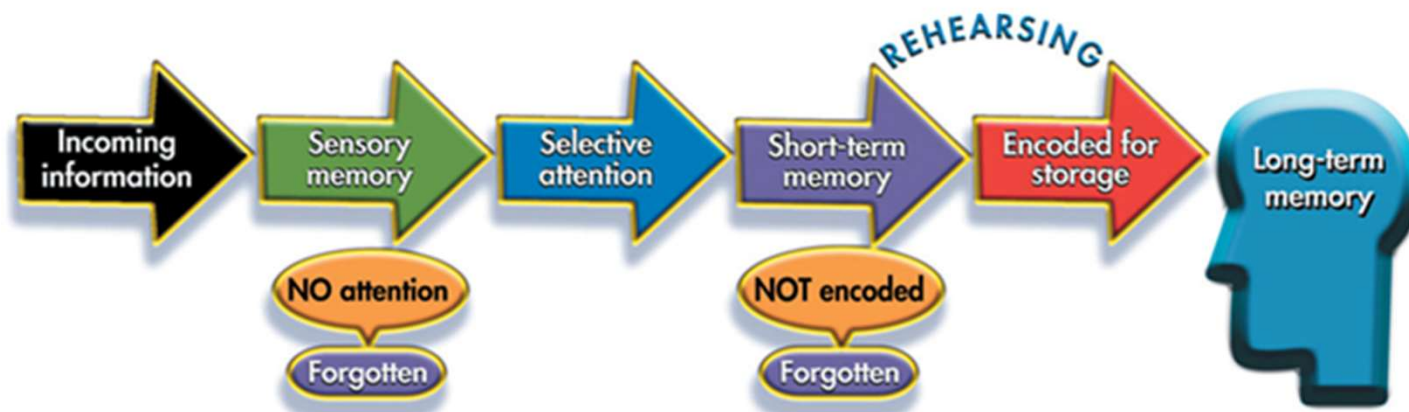
–Short-term memory

don't pay attention, information isn't encoded and is forgotten

–Long-term memory

encoded information will remain on a relatively permanent basis

THREE TYPES OF MEMORY (cont.)



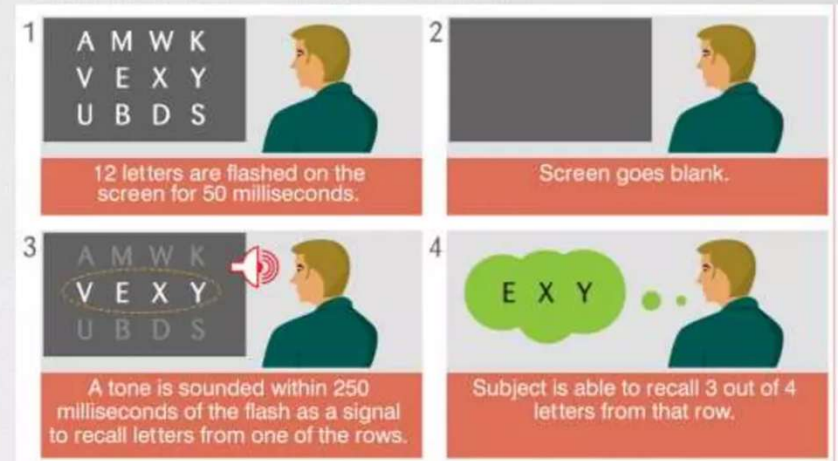
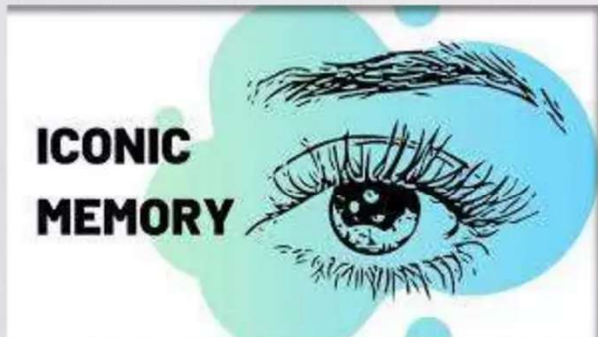
(c)2011 Cengage Learning

SENSORY MEMORY

We receive sensation from the sensory modules. visual and auditory system based on memory processes have been studied most

ICONIC ((Visual)

Holds information for 1-2secs and 11-16 items. Can hold ftures such as- colour, shape, and direction.

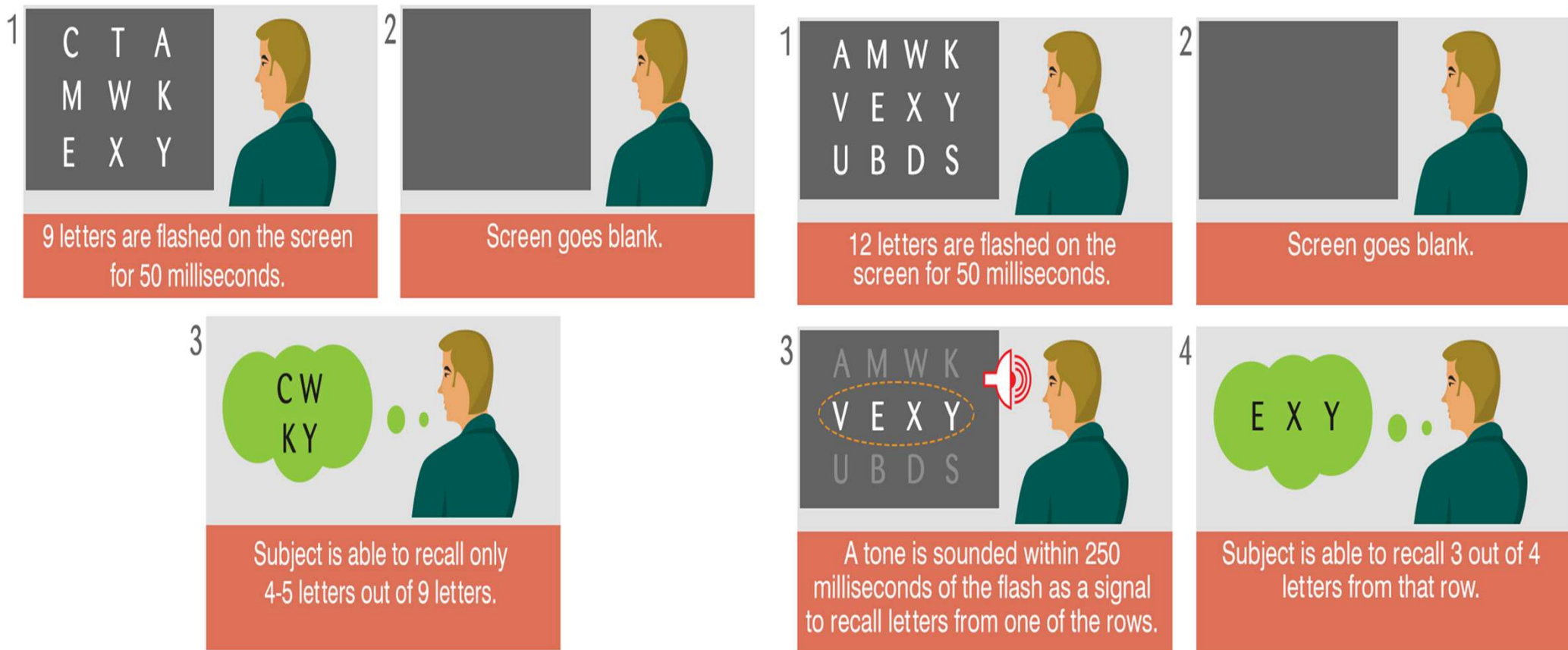


ECHOIC (Auditory)

Holds information for up to about 4-5 secs. can store sequences of at least 250 milisecs or more



George Sperling's Experiment:



SENSORY MEMORY: RECORDING

Iconic memory –Form of sensory memory that automatically holds visual information for about a quarter of a second or more; as soon as you shift your attention, the information disappears

–Icon means image

Echoic memory

–Form of sensory memory that holds auditory information for 1 to 2 seconds

–Holds speech sounds long enough to know that sequences of certain sounds from words

SENSORY MEMORY: RECORDING (CONT'D)

Functions of sensory memory

- Prevents being overwhelmed
- Gives decision time
- Provides stability, playback, and recognition

Activity : sperling's experiment-

SPEERLING EXPERIMENT

Display 3-16 characters for short period of time

Memory performance was compared under whole and partial report condition.

WHOLE- recall as many as displayed elemented as possible. Irrespective of characters displayed , 4-5 characters were recollected.

Partial- Identification of the subset of the characteristics within the visual display.

RESULT-

Reported characters are indicated with a glow, cued row indicated by an arrow.

Whole report: subjects are presented with a matrix of alphanumeric characters for 15-500ms. When asked to report all of the characters they can remember, subjects can report on average 4-5 /12 characters, an accuracy of 33-40%.



Partial report: However, if subjects are cued to report just on *one particular row* (top, middle, or bottom) subjects can correctly remember 3-4 characters from that row, an accuracy of 75-100%, but few if any others. This effect persisted for up to 1000ms in Sperling's original experiment.



SHORT-TERM MEMORY: WORKING

Short-term or working memory

- Process of **holding a limited amount of information** (an average of seven items) for a limited period of time.(2 to 30 seconds)
- Short term memory can be extended through repetition or rehearsal of the information.

Two features

- Limited duration
- Maintenance rehearsal practice of intentionally repeating information so that it remains in short-term memory longer

SHORT-TERM MEMORY: WORKING (CONT'D)

Interference

–Results when new information enters short-term memory and **overwrites** or pushes out information that's already there

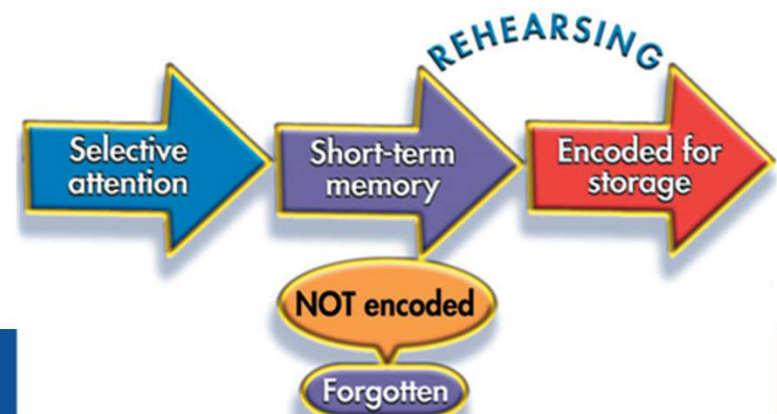
Chunking

–Combining separate items of information into a larger unit, or chunk, and then **remembering these chunks rather than individual items**

SHORT-TERM MEMORY: WORKING (CONT'D)

Functions of short-term memory

- Attending** selectively attend to relevant information and disregard everything else
- Rehearsing** allows you to hold information for a short period of time until you decide what to do with it
- Storing** helps store or encode information in long-term memory



Rehearsal in STM

```
graph TD; A[Rehearsal in STM] --> B[Maintenance]; A --> C[Elaborative];
```

Maintenance

Repetition of information over and over again . It does not necessarily succeed transferring the information into LTM.

Elaborative

Organizing the given material and giving meaning to it is being rehearsal. This helps in fitting it with the existing organized LTM.

7 to 2 chunks of memory can be kept in mind at a time and combined to make a decision. The size of the chunk varies greatly between experts and novices. Experts can combine many memory units to a whole .

LONG TERM MEMORY (LTM)

Stores information over a long time. It has unlimited capacity. Informations which are rehearsed in the STM are transferred into LTM.



- Declarative: memory for facts!



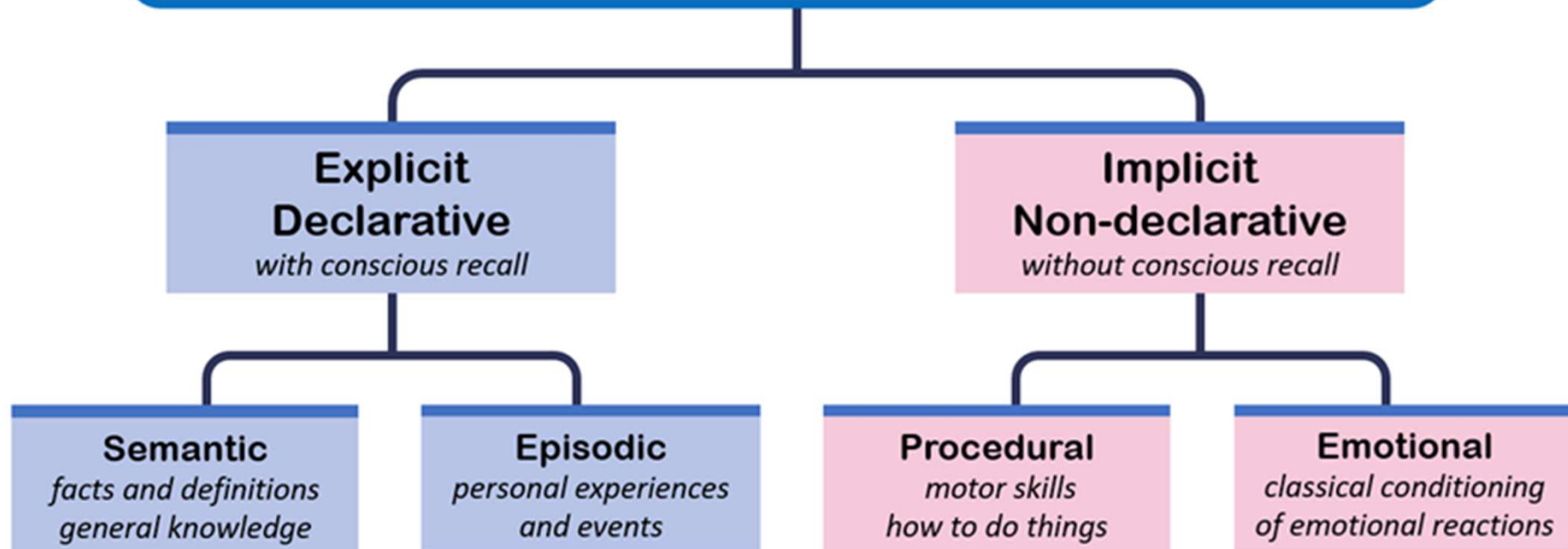
- Procedural: memory for skills and procedure.



Declarative Memory Is Further Broken Down:

- Semantic : words, concepts, information that can be described and applied.
- Episodic : personal experiences and events.

Types of Long-Term Memory



(Based on Malmberg et al., 2019)

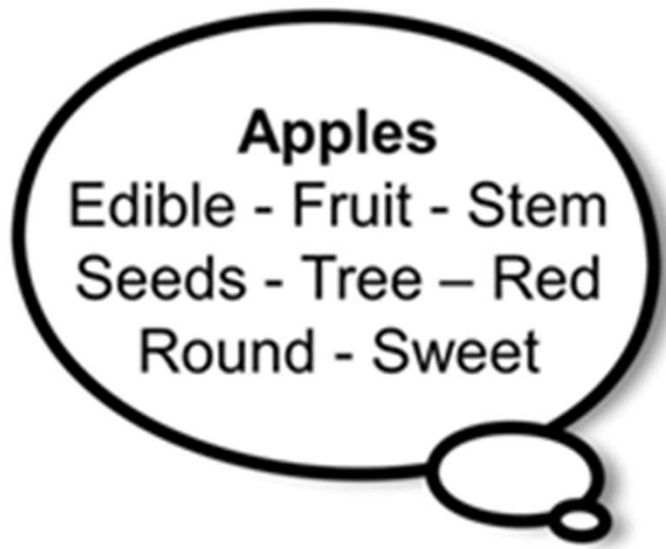
Explicit Memory

- Encoded to memory and later retrieved
- Often formed deliberately through rehearsal
- Often encoded unconsciously and tied to emotions
- May be drawn into awareness through associations

Implicit Memory

- Becomes automatic over time with repetition
- Begins with learning skills and mastering a task
- Can result in priming, or responding the same way to similar stimuli
- Often dependent upon context and cues

Semantic Memory



object knowledge learned
over many interactions

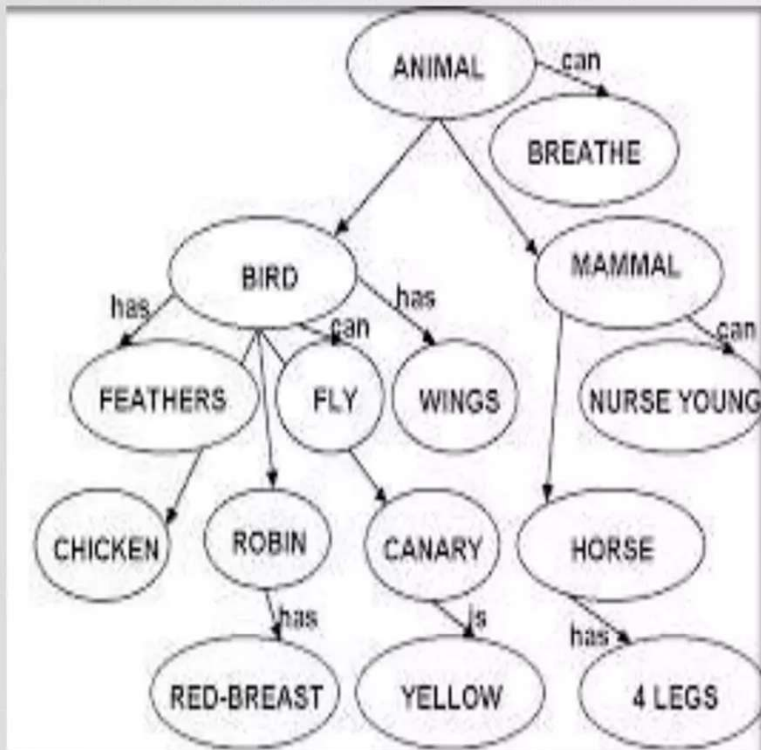
Episodic Memory



memory for specific events
that you have experienced



ORGANIZATIONS OF SEMANTIC MEMORY



SEMANTIC NETWORK-It is the knowledge representing meaningful relationship among concepts.

CONCEPTS- Mental categories that are used in the organization of objects or events.

PROTOTYPES- They best represent the categories. They are the best examples of any given concept that readily comes to one's mind.

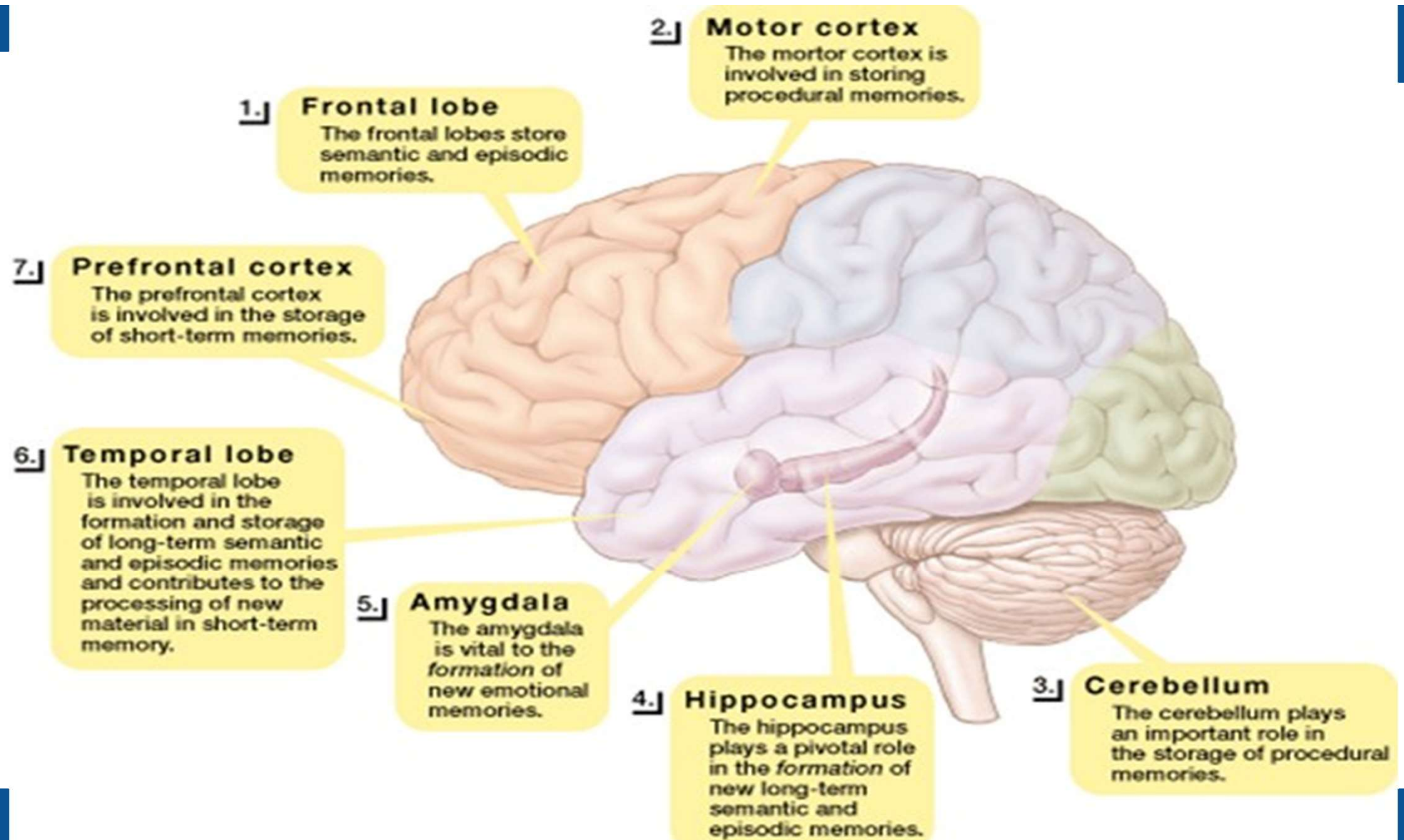
Episodic memories

Episodic memories usually include details of an event, the context in which the event took place, and emotions associated with the event.

It involves conscious thought and is declarative.

Your memories of your first day of school, and your graduation are all examples of episodic memories.





Implicit memories/ non declarative :

A type of long-term memory that a person remembers unconsciously, unintentionally, and effortlessly, such as certain abilities or skills.

They are usually non-conscious and not verbally articulated, often procedural and focused on the step-by-step processes that must be performed in order to complete a task.

But they still influence how one behaves as well as one's knowledge of different tasks.



Procedural memories: example

- singing a familiar song.
- reading a book
- typing.
- writing, painting
- brushing teeth
- boiling water, cooking

Procedural Memory: Habits and skill. Recalling motor and executive skills.



Emotional conditioning:

Events that cause strong emotions can condition people to recognize stimuli quickly and react accordingly.

“Emotional memory” is shorthand for denoting the memory of experiences that evoked an emotional reaction.

It is most commonly used to refer to the ability to consciously remember aspects of those experiences.

Eg : A good example of this is people who are afraid of thunder



OTHER FORMS OF MEMORY



EYEWITNESS MEMORY

A form of episodic memory pertaining to specific events. It is for forensic importance. Vulnerable to distortion. Visual recall of events.



AUTOBIOGRAPHICAL MEMORY

Memories about events and experiences of one's own lives. Recollections of specific life events.



FLASHBULB MEMORY

Specific mechanism hypothesis (Brown and Kulik, 1977). Specific memory of events with extremely high level of surprise and consequences attached to it. One type of autobiographical memory.



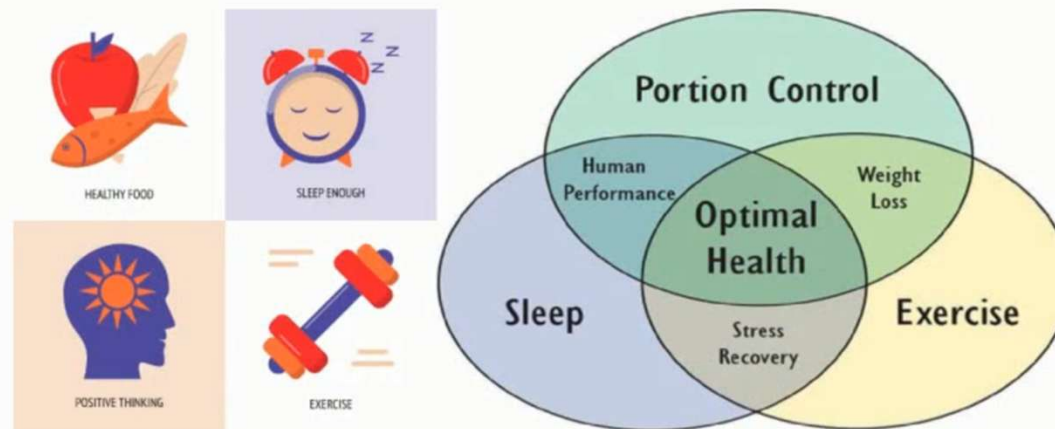
Factors influencing memory



(Factors to influence the memories)

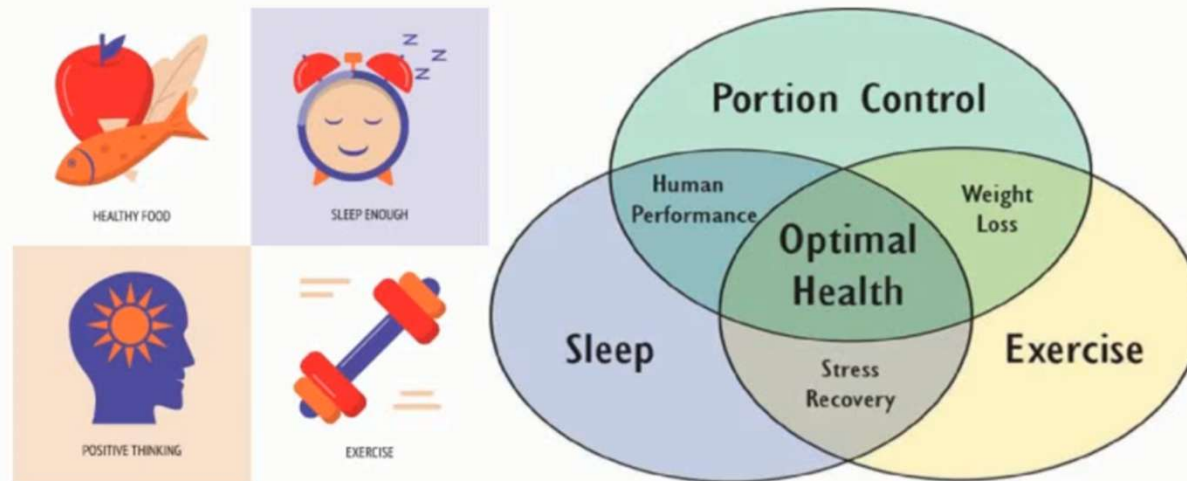
- stress: Facilitate the formation of implicit memories for negative emotional information.
- mood: People experiencing depressed moods are more likely to recall negative information implicitly, while not experiencing depressed moods, positive information implicitly.

Ways to protect memory



(Ways to protect the memories)

- sleep: Enough sleep plays an essential role in the memory consolidation process.
 - physical exercise: Good for both the brain and body.
- mental exercises: Brain training activities can help keep the mind sharp and may even help strengthen certain cognitive skills.



- healthy diet: Foods that are good for the brain include fruits, vegetables, fatty fish, nuts, and eggs.

Ref for long term memory :

<https://www.frontiersin.org/journals/pharmacology/articles/10.3389/fphar.2017.00438/full>

Comparison of Three Stages of Memory

Sensory

1. Large capacity
2. Contains sensory information
3. Very brief retention (1/2 sec for visual; 2 secs for auditory)

Short Term

1. Limited capacity
2. Acoustically encoded
3. Brief storage (up to 30 seconds w/o rehearsal)
4. Conscious processing of information

Long Term

1. Unlimited capacity
2. Semantically encoded
3. Storage presumed permanent
4. Information highly organized

COMPARING MEMORY TYPES

TOPIC	SENSORY	STM	LTM
DURATION	ICONIC - 1-2 Secs ECHOIC- 4-5 Secs	20-30 Secs	Days, months, years or life time
CAPACITY	Relatively large, at least 16 items, probably much more	Relatively small. About 7 items or chunks	Very large limit unknown
TRANSFERENCE	Attention and recognition . Items attended to and recognized move to STM.	Items approx. rehearsed move to LTM.	N/A
TYPE OF INFORMATION STORED	Copy of input	Sounds, visual image, words and sentences	Primarily meaningful sentences, life events and concepts, some images semantic and episodic memory
REASON OF INFORMATION LOSS	Decay of trace	Displacement of old information by new information	Faulty organization inappropriate interference

Case study

<https://youtu.be/KkaXNvzE4pk>

Factors influencing memory performance

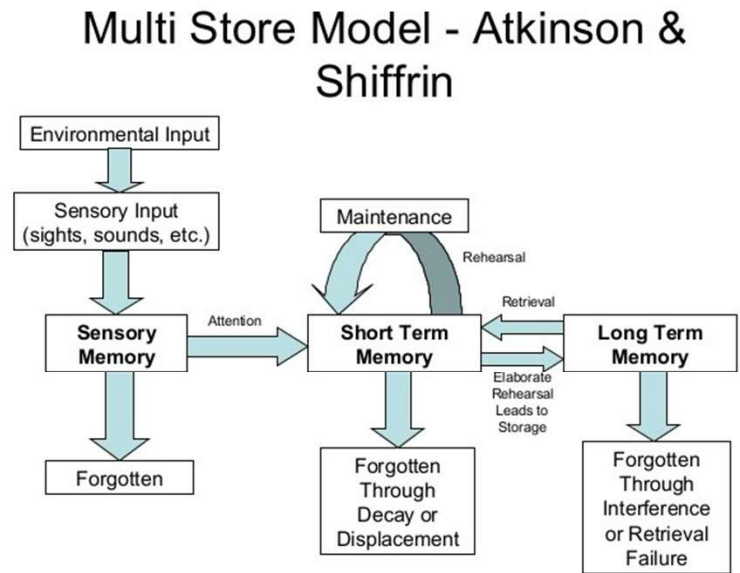
Activity: Discuss and present the Factors influencing memory performance

15 min for preparation : 5 min for presentation

Ref material : <https://medium.com/@helenjoy88/factors-influencing-memory-performance-oh-really-aa3e5001d13a>

Memory models

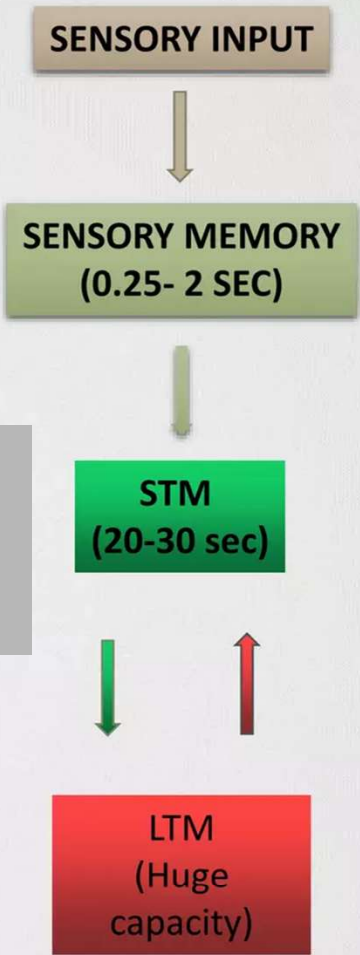
ATKINSON AND SHIFFRIN'S MODEL (1968)



Some materials which are given attention transferred to STM

Some informations are rehearsed and transformed to LTM

RETRIEVAL



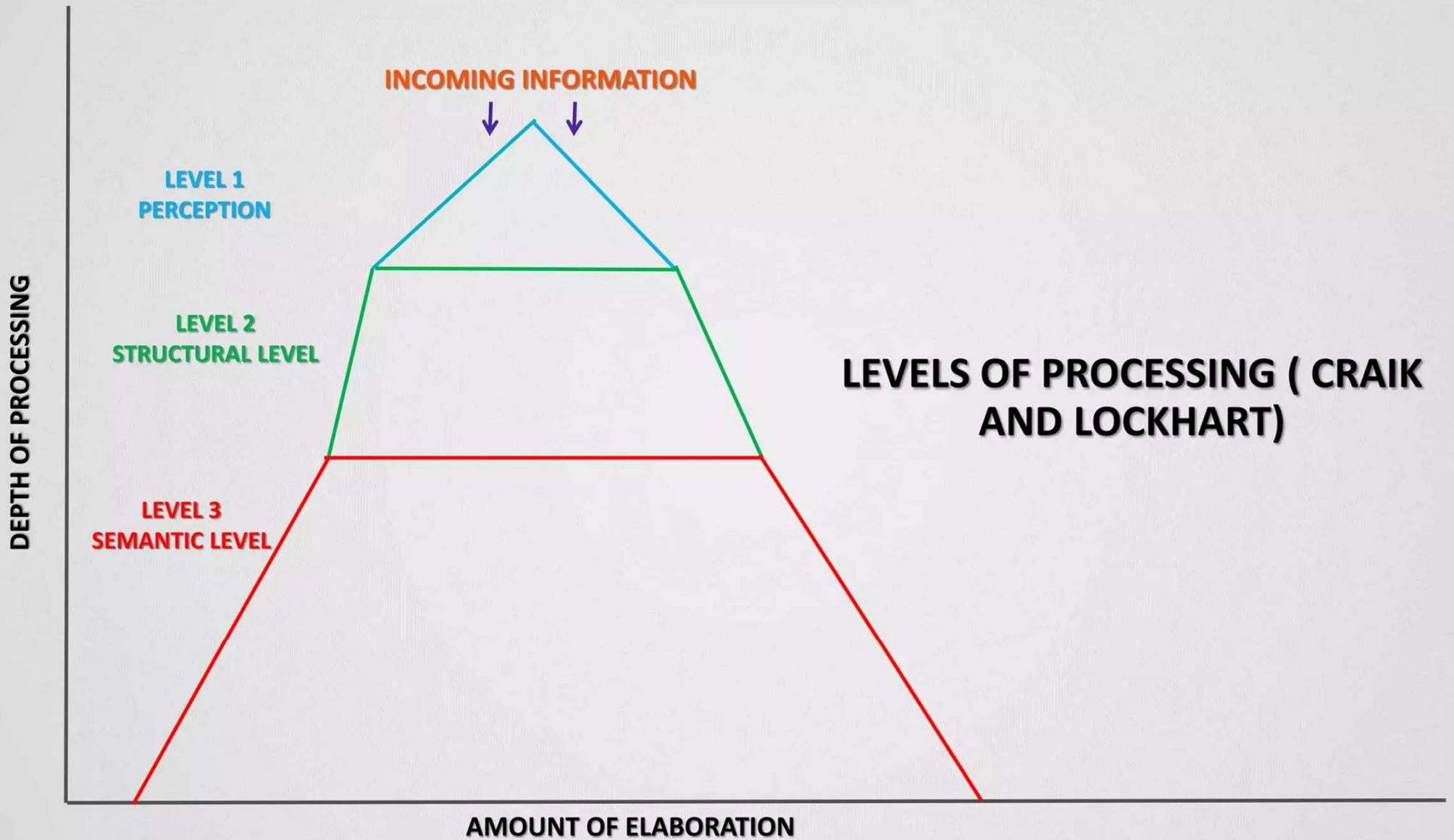
Some informations are not transferred to STM and is forgotten



Some informations are rehearsed are forgotten

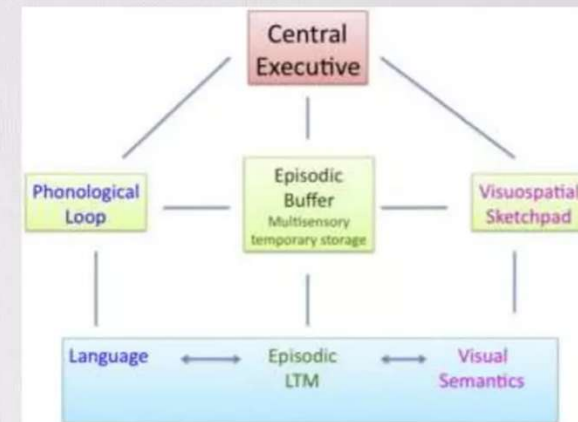
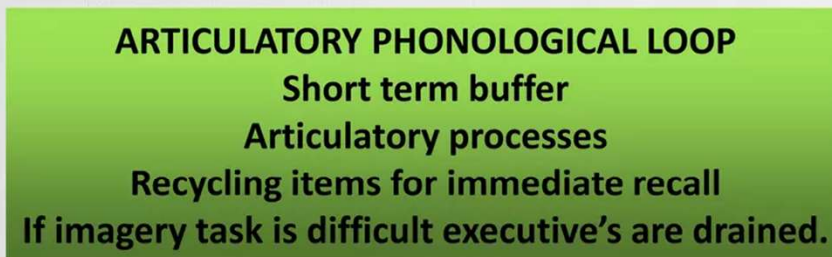
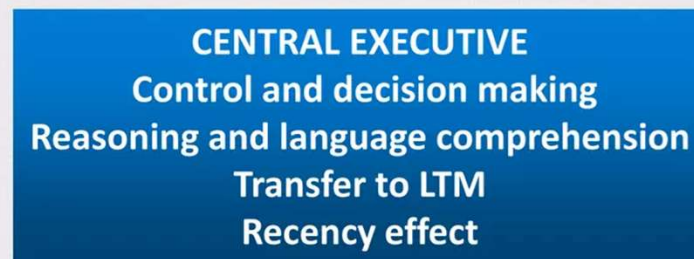


Factors like decay and inference may lead to forgetting from LTM



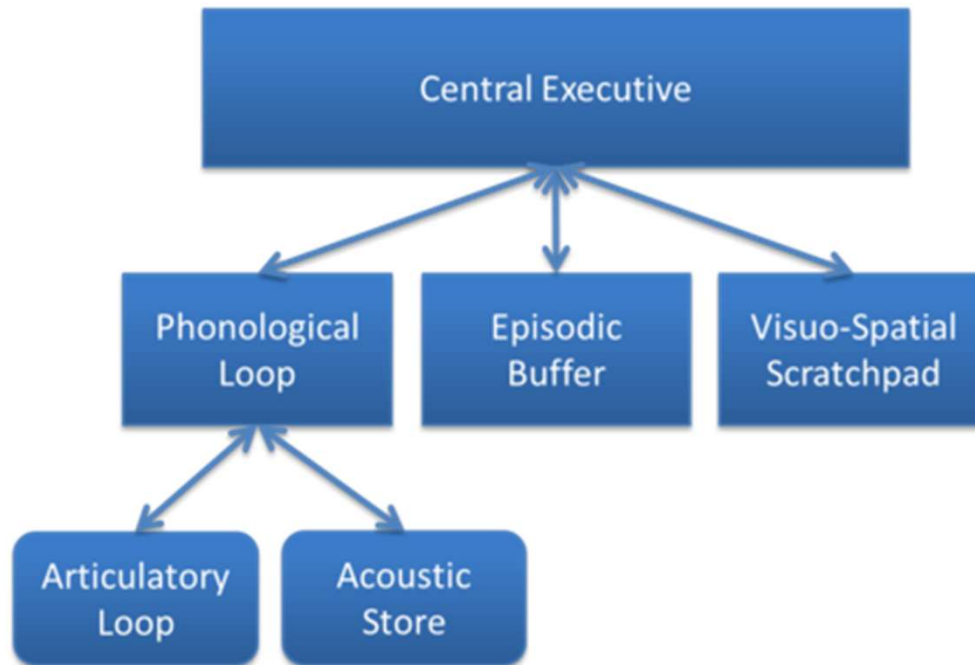
ALLEN D. BADDELEY'S MODEL

- ✓ **ARTICULATORY LOOP**- Site of verbal rehearsal
- ✓ **VISUAL -SPATIAL SKETCHPAD**-Site of rehearsal of visually presented information.

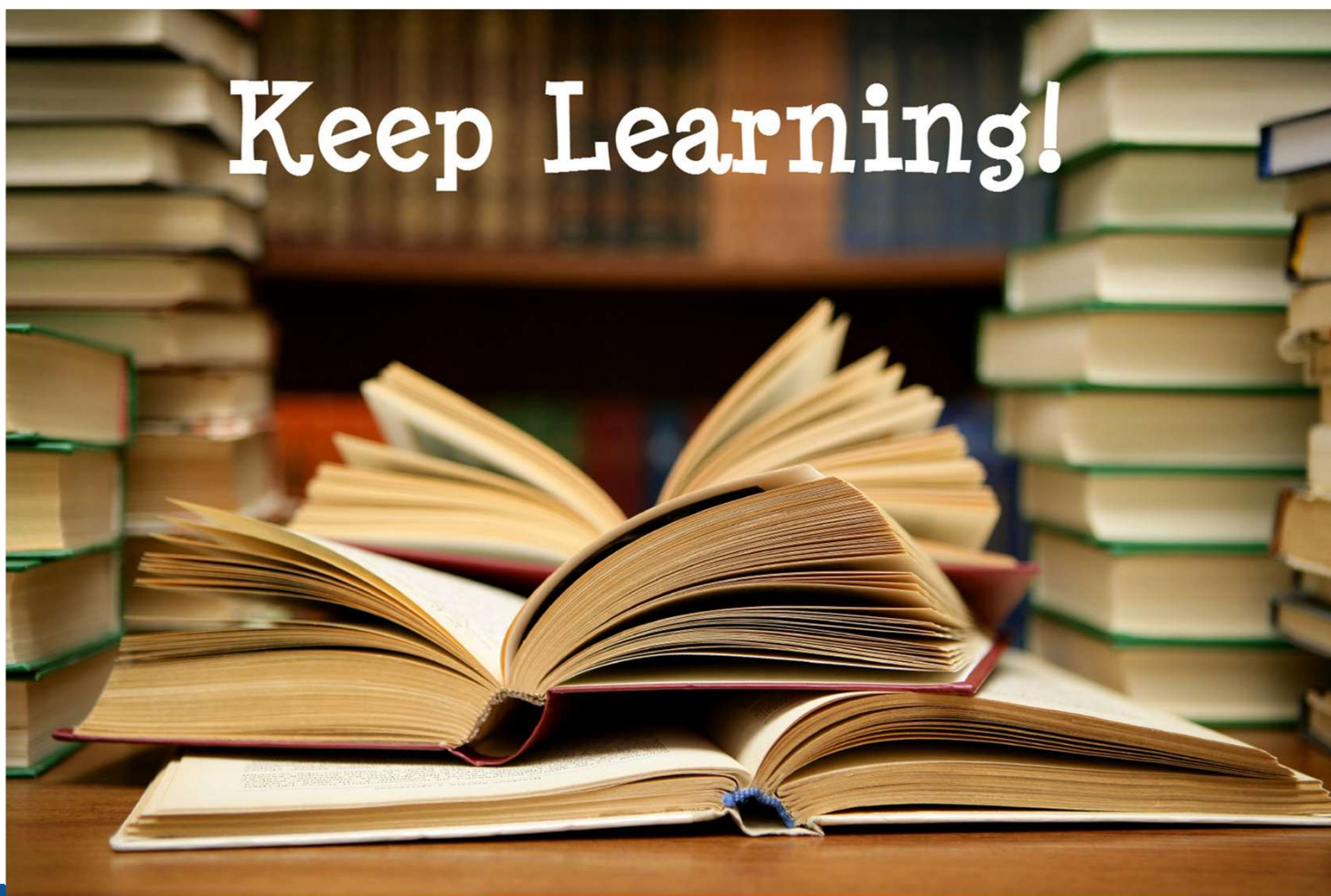


Baddeley's model of working memory

The Working Memory Model



https://en.wikipedia.org/wiki/Baddeley%27s_model_of_working_memory



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